

The Ontological Status of Time

Coordinate as Bookkeeping Parameter, Not the Fourth Dimension

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Abstract

This paper distinguishes mathematical structure, physical representation, and physical ontology. A mathematical structure can describe physical phenomena with complete predictive accuracy and still acquire no ontological status: description is not identity, and predictive success is not ontological licensing. These principles are systematically violated in the standard interpretation of four-dimensional spacetime, and that violation is what this paper corrects.

The paper's position is stated without qualification: time is not a dimension. Time is an irreversible ordering relation of states. The coordinate t in the equations of special and general relativity is a bookkeeping parameter—an indexing device for ordering events—not time itself. The four-dimensional spacetime manifold, offered as a description of physical reality, is a successful geometric representation of event relations. It is not a physical ontology of time. The paper does not contest the empirical predictions of relativity theory; the Lorentz transformations, the Minkowski metric, and the geodesic equations remain correct. What is contested is the inferential move from " t is a useful mathematical parameter" to "time is therefore a geometric dimension ontologically parallel to the three spatial dimensions." This move is not licensed by the mathematics; it is an additional philosophical assumption, and this paper argues that the assumption is false.

The argument proceeds in four stages. A structural analysis shows the directional asymmetry of time is not the kind of thing a geometric dimension can carry. The algebraic structure appropriate to time—a semigroup ordering relation—is shown categorically different from the group structure of spatial dimensions. A thought experiment separates the coordinate t from temporal ontology. Standard defences of the four-dimensional ontology are shown to be self-undermining. The conclusion: t must be demoted from the world's fourth axis to the description's fourth column.

Keywords: philosophy of time; ontology of spacetime; four-dimensionalism; block universe; semigroup structure; temporal asymmetry; coordinate time; category error

Section 1: The Problem—The One-Way Wall

1.1 Thought Experiment One: The Wall

Consider a being who enters a region of space that has been described to them as possessing four dimensions: three spatial and one temporal, with the temporal dimension running in both positive and negative directions, just as the spatial dimensions do. In the three spatial dimensions, the being moves freely. They walk east and then west. They ascend and then descend. They move forward and then backward. In each spatial case, the reversal costs energy and requires effort, but it is structurally available—there is no absolute barrier, only physical resistance.

Now the being attempts to move along the temporal dimension in its negative direction: to return to yesterday. They do not encounter friction. They do not encounter an energy threshold that exceeds their capacity. They encounter a wall. The wall is not a probabilistic obstacle—it is not the case that the probability of returning to yesterday is very low, or that the entropy cost is prohibitive, or that the available information is insufficient for navigation. The wall is structural. The negative temporal direction does not merely resist traversal; it does not exist as a traversable direction at all. The being pushes against it and finds not resistance but absence.

This thought experiment is not about the empirical difficulty of time travel. It is a conceptual probe designed to isolate a structural property. The question it poses is precise: if a direction that is called a "dimension" has the property that traversal in one of its two orientations is structurally unavailable rather than merely practically difficult, what does this reveal about the nature of that direction?

The answer is immediate and unambiguous. A spatial dimension, in the sense in which the three spatial dimensions are genuinely spatial, is structured as a bidirectional, group-like axis: traversal in either orientation is defined, and every displacement has a structural inverse. The temporal direction lacks this structure. East and west are both available. Up and down are both available. Positive x and negative x are both available. A direction that permits traversal in only one orientation is not a dimension in this sense. It is an ordering relation—a structure that assigns a sequence to states without providing a path back through that sequence. The wall is not evidence that time travel is hard. The wall is evidence that the negative temporal direction is not a direction in the geometric sense. It is the absence of a direction where the four-dimensional picture placed one.

Two clarifications must be made explicit before proceeding. First, this paper does not discuss entropy, thermodynamic time arrows, or statistical irreversibility. These are macroscopic emergence phenomena and the irreversibility they describe has a well-understood explanation: coarse-graining over microscopic degrees of freedom produces effective asymmetry at the macroscopic level. The wall in the thought experiment is not this kind of asymmetry. Statistical asymmetry means that the reverse process is improbable. Structural asymmetry means that the reverse process is undefined [1,2]. The being in the thought experiment is not facing long odds. They are facing a logical absence. Second, the thought experiment does not depend on any particular physical theory. It is a conceptual instrument that can be applied prior to and independently of the formalism of relativity, quantum mechanics, or any other theory. The question it raises—what kind of structure can generate absolute directional asymmetry?—must be answered before any of those formalisms can legitimately claim to have incorporated time into their framework.

1.2 Four Possible Sources of the Wall—Exhaustive Analysis and Verdict

The wall must have a source. Four candidate sources present themselves, and they must be examined exhaustively before any conclusion about the ontological status of time can be drawn. The argument of

this paper is that all four candidates, when followed to their conclusions, converge on the same result: time is not a geometric dimension of the same kind as space.

Possibility One: Boundary Condition.

Yesterday is not an accessible region; it is only a mathematically extended coordinate value—like the "territory beyond the cliff edge" that the map represents but that no agent can reach by walking. On this account, the wall is a boundary condition imposed on the geometry: the physical world has a one-sided extent in the temporal direction, and the negative half of the t-axis is a mathematical artefact with no physical territory behind it.

Verdict: If this possibility is correct, the negative half of the t-axis is an empty label. A coordinate axis of which one half is physically empty does not describe a dimension in any ontologically robust sense. It describes a half-line—a ray—dressed up in dimensional language. The implications of this verdict are developed in Section 4.

Possibility Two: Information Loss.

The state of the world today does not contain sufficient information to uniquely reconstruct the state of the world yesterday. Returning to yesterday is not merely difficult; it is undefined, because the present state is not a unique image of the past state under any recoverable mapping. The mapping from earlier realised states to later realised states is non-injective in the physically accessible domain: the information required to uniquely reconstruct the prior state is not preserved in the current state description. Many past states can give rise to the same present state, so no inverse mapping exists. On this account, the wall is a consequence of information collapse.

Verdict: If the fundamental evolution of the world is non-injective—if states are compressed rather than preserved—then the temporal "dimension" is not a dimension at all, but a collapsing funnel. The algebraic structure this implies is examined in Section 2.

Possibility Three: Time is Not a Dimension but an Irreversible Ordering Relation.

There is no traversable negative temporal direction because time does not consist of a traversable axis in either direction. Time is the irreversible ordering relation of states: state A is followed by state B, and this relation of following has no inverse definition. The ordering exists; the reversal of the ordering does not. The wall is not a barrier in front of a destination that exists; it is the structural fact that there is no destination, because the relation that would have to connect the present to the past in the reverse direction simply has no definition.

Verdict: This paper takes this to be the correct characterisation of temporal structure. Possibilities One and Two, when fully analysed, collapse into this characterisation: the boundary condition of Possibility One is itself a consequence of the absence of a reverse-ordering structure, and the information loss of Possibility Two is itself a consequence of the fact that the state-ordering relation is non-invertible. Possibility Three is not one candidate among others; it is the structural ground from which the other two possibilities emerge [3,4]. Crucially, this paper does not further specify the ontological residence of this ordering relation—whether it belongs to the universe as a whole, to information structures, or to observers—because that question is beyond the scope of the present argument and will be addressed in subsequent work (see Section 9.6).

Possibility Four: A Negative-Time Domain Exists.

The wall is real and the region behind it is also real—a physical domain in which the temporal coordinate takes negative values, constituting a mirror spacetime simultaneous with ours but inaccessible to observers in our domain. On this account, four-dimensional time is vindicated: t runs

from negative infinity to positive infinity, both halves are physically real, and the wall is merely an accessibility constraint specific to our domain.

Verdict: This possibility cannot be absorbed into Possibility Three; it must be handled separately because it is the strongest available defence of the four-dimensional ontology. But the defence collapses into a dilemma. Accepting Possibility Four requires accepting that the observable universe is not the only spacetime—that there exists a physically real negative-time domain alongside it. This is an empirically unbounded commitment that standard physical practice categorically refuses to make. The moment a physical theory posits a physically real domain that is in principle inaccessible to all observers in our spacetime, it has exceeded the boundary of empirically constrained theorising and entered speculative metaphysics. If the defenders of the four-dimensional ontology accept this cost, they have conceded the game to metaphysics. If they reject Possibility Four—as empirical practice demands—they concede that the negative half of the t -axis has no physical referent, which places them directly in the second horn of the dilemma developed in Section 4. Possibility Four is the last escape route for the four-dimensional ontology. It is a route that leads directly back into the trap.

Convergence.

All four possibilities, when traced to their structural conclusions, produce the same verdict: time does not function as a geometric dimension of the same kind as the three spatial dimensions. This is an exhaustive argument. No fifth possibility has been identified, and none is available, because the four candidates above span the logical space: the wall is either a boundary condition, or an information constraint, or a consequence of the non-existence of a reverse-ordering relation, or evidence of a real but inaccessible negative domain—and in every case, the four-dimensional ontology fails.

Section 2: Mathematical Structure—Semigroup Against Group

2.1 Reversibility: Precise Mathematical Definition

Before any claim about the algebraic structure of time can be evaluated, the relevant mathematical concepts must be stated precisely, because the most common errors in this domain arise from conflating three distinct situations that have different mathematical characters.

A function f from a set X to a set Y is invertible if and only if it is a bijection—a mapping that is both injective (one-to-one) and surjective (onto). Invertibility means that there exists a function g such that $g(f(x)) = x$ for all x in X , and $f(g(y)) = y$ for all y in Y . This is the condition of full reversibility: the original information is completely preserved, and the function can be run in both directions without loss.

Two weaker situations must be distinguished from genuine irreversibility, because they are frequently misdescribed as irreversibility when they are not.

The first is non-unique invertibility. Consider $f(x) = x^2$. This function is not bijective on the real numbers: both $x = 2$ and $x = -2$ map to 4. The function is not invertible, but this is not irreversibility in the structurally significant sense. The original state ($x = 2$ or $x = -2$) is not lost; it is merely ambiguous given only the output. Both original states are mathematically available. An observer who knows additional information—for example, that the original x was positive—can recover the unique original state. The failure of invertibility here is a failure of unique recovery, not a failure of existence of the original.

The second is genuine information destruction. Consider $f(x) = 0$ for all x in X . Every element of X —whether $x = 1$, $x = 100$, or $x = -7$ —maps to the same output, 0. From the output, no recovery of the original is possible, not because recovery is ambiguous but because no information distinguishing the original states survives in the output. There exists no function g such that $g(f(x)) = x$ for all x , because

$g(0)$ cannot simultaneously equal 1 and 100 and -7. This is structural irreversibility: the mapping has destroyed the information that would be required for any inverse to exist.

The dimensional projection provides the clearest spatial illustration. The map $(x, y) \rightarrow (x, 0)$ takes every point in the plane and eliminates its y -coordinate. The output $(5, 0)$ could have come from $(5, 1)$, $(5, 100)$, $(5, -999)$, or any point of the form $(5, y)$. The inverse does not exist—not because we lack the computational capacity to run it, but because the information distinguishing these origins has been structurally eliminated. No amount of additional computation recovers what the projection has erased, because the information is not hidden; it is absent.

These distinctions matter because the standard defence of reversible physical evolution relies on conflating genuine irreversibility with non-unique invertibility. When a physicist says "the Schrödinger evolution is reversible," they mean that $U(t)$ is a bijection on the Hilbert space [5]—the evolution is injective and surjective, and the inverse $U(-t)$ exists and is also unitary. This is correct as a statement about the mathematical structure of the evolution operator. It becomes incorrect the moment it is used to claim that the temporal process described by that operator is therefore reversible in the ontological sense—that is, that actual states can return to actual prior states. The mathematical existence of $U(-t)$ does not guarantee the physical existence of a world-state that implements $U(-t)$. This distinction is the load-bearing element of Section 2.4.

2.2 Group and Semigroup: The Ontological Significance

The algebraic structures of groups and semigroups are not merely technical classifications. They encode fundamentally different kinds of physical reality.

A group is a set G equipped with a binary operation $\cdot$ that satisfies four axioms: closure (for all a, b in G , $a \cdot b$ is in G), associativity, the existence of an identity element e such that $a \cdot e = e \cdot a = a$ for all a , and the existence of an inverse a^{-1} for every element a such that $a \cdot a^{-1} = a^{-1} \cdot a = e$. The existence of inverses is the defining feature that makes group structure equivalent to traversable structure: every operation has an undoing, every step has a return.

Spatial translations form a group. The operation of translating by vector v has the inverse of translating by $-v$. The operation of rotating by angle θ has the inverse of rotating by $-\theta$. The symmetry group of spatial translations is precisely the mathematical expression of the fact that space is traversable in both directions: whatever motion is possible in one direction, the reverse motion is equally possible.

A semigroup is a set S equipped with a binary operation $\cdot$ that satisfies only closure and associativity. It requires neither an identity element nor inverses. A semigroup operation has a "next step" but not necessarily a "return step." The sequence of states $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3$ has the associative composition property—applying the $S_0 \rightarrow S_1$ transition followed by the $S_1 \rightarrow S_2$ transition gives the same result as applying a composed $S_0 \rightarrow S_2$ transition—but the reverse transitions $S_1 \rightarrow S_0$, $S_2 \rightarrow S_1$ need not exist.

The core claim of this paper regarding algebraic structure is this: spatial structure is group-like—bidirectional, with inverses defined for all displacements—while the structure of realised temporal succession is semigroup-like: the forward ordering is defined, the inverse is not. These are categorically different structures, not merely quantitatively different degrees of the same kind. The difference is not that time is "more difficult" to traverse in reverse than space. The difference is that the reverse operation exists in the group structure of space and does not exist in the semigroup structure of time. Embedding a semigroup object into a group-structured geometric framework and asserting that they are the same kind of thing is not a simplification or an approximation. It is a category error.

The standard physical theories—Newtonian mechanics, Schrödinger quantum mechanics, general relativistic field equations—all employ time-reversal symmetric formulations [6,7]. T -symmetry is a mathematical property of these equations: if a solution exists for the sequence $S_0 \rightarrow S_1 \rightarrow S_2$, then a solution also exists for $S_2 \rightarrow S_1 \rightarrow S_0$. This is a fact about the mathematical structure of these theories. The claim of this paper is that this fact reflects a deliberate theoretical choice—the choice to work with

information-preserving, group-structured evolution equations—and not a fact about time itself. Physical theories prefer reversible mathematical structures because reversible structures are computationally complete and analytically tractable. The preference for group-structured evolution is a methodological choice. Reading the properties of the methodological choice back onto the object being described—concluding that because the theory is time-reversal symmetric, time itself is group-structured—is the category error that this paper targets.

2.3 The Completeness Clause

The analysis of algebraic structure generates an immediate consequence for the four-dimensional spacetime framework. The argument takes the following form.

If a theory claims to describe a closed and informationally complete state space—one that omits no physically relevant degrees of freedom—then its fundamental evolution must be a bijection; if the evolution is not a bijection, information is being lost from the claimed complete description. If the evolution is not a bijection, then either state information is being discarded (the evolution is a compression, not a description of reality) or the state space is not complete (the theory is not describing everything that is physically relevant). This is a general logical constraint on any physical theory that claims completeness.

Four-dimensional spacetime claims to be a complete geometric container for physical events. Every event is assigned coordinates (t, x, y, z) , and the geometric structure of the manifold describes the relations among events. The claim is not merely that this is a useful parameterisation; the four-dimensional ontology asserts that the manifold itself is the physical reality—that the events are "already there," laid out across all of space and time in a block that exists timelessly.

But four-dimensional spacetime, despite accommodating both the positive and negative directions of the t -axis in its mathematical structure, cannot derive the unidirectionality of time from within its geometric structure. The Minkowski metric $ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$ assigns a negative signature to the temporal direction, which distinguishes timelike from spacelike separations and allows the construction of light cones. But the metric signature does not generate directional asymmetry within the timelike direction. Both future-directed and past-directed timelike vectors exist in Minkowski geometry; the metric assigns them equal standing. The asymmetry between past and future—the wall—is not a consequence of the metric. It is a fact the metric cannot see.

A theory that claims to be a complete description of space and time, but cannot derive from within its own structure the most fundamental structural feature of time—its unidirectionality—is not a complete description. It accommodates both sides of the wall but cannot explain the wall. A framework that contains what it cannot explain does not earn the title of complete description by the containing. The incompleteness charge is not that four-dimensional spacetime is wrong in its predictions. The incompleteness charge is that it fails the criterion a complete ontological description would have to meet: it would have to generate, from within its own structure, the constraint that defines what time is.

2.4 Direct Rebuttal: Against the Claim That Evolution Operators Form a Group

The most technically serious objection to the semigroup characterisation of temporal structure is this: standard quantum mechanics represents time evolution by the one-parameter unitary group $U(t)$, which is fully invertible. $U(-t) = U(t)^\dagger$ exists, is unitary, and maps any state back to its prior state with complete fidelity. Semigroup structure appears only in open systems, dissipative dynamics, and coarse-grained effective descriptions (the Lindblad equation being the standard example). Therefore, the semigroup characterisation of time is not a claim about the fundamental structure of temporal evolution; it is a feature of a restricted and non-fundamental domain.

This objection conflates three distinctions that must be kept separate.

First: the distinction between the description's properties and the described object's properties.

$U(t)$ is a mapping between descriptions of states—specifically, a mapping between vectors in a Hilbert space that represents the probability structure of possible physical configurations. The invertibility of $U(t)$ is a property of this mapping. It means that the mathematical structure of quantum descriptions is informationally closed: no information about the description of a state is lost under the evolution of the description. This says nothing, directly, about whether the physical states being described are reversible. A double-entry bookkeeping system is perfectly reversible: every ledger entry can be traced, every transaction reversed on paper. This does not mean that the economic transactions themselves are reversible. The reversibility of the accounting system is a property of the accounting system. $U(t)$'s reversibility is a property of the quantum-mechanical accounting system for physical states. The two levels must not be conflated.

Second: the distinction between the space of possible evolutions and the sequence of realised states.

The inverse operator $U(-t)$ exists in the mathematical structure of quantum mechanics. Its existence means that the reverse evolution is a mathematically well-defined object in the space of possible descriptions. It does not mean that the reverse evolution is physically realised in the sequence of actual world-states. No segment of the universe's actual history has ever been identified as the time-reverse of another segment. The group structure of $U(t)$ lives in the space of possible descriptions; the irreversible ordering relation lives in the sequence of realised states. These are different domains, and the properties of one do not transfer automatically to the other.

This distinction is not novel to the present paper. It is the temporal instance of the ontological boundary identified in the author's previously published framework for the interface between quantum mechanics and general relativity [8]. The staged mapping $|\psi\rangle \rightarrow C_i \rightarrow T_{\mu\nu}(C_i) \rightarrow G_{\mu\nu}$ provides a formal expression of the boundary between the domain of quantum probability structures and the domain of physically actualised facts [9]. Prior to collapse C_i , the quantum state $|\psi\rangle$ belongs to the space of possible configurations—a space that has exactly the group structure, the unitarity, and the reversibility of the Schrödinger evolution. After collapse, the realised physical fact C_i belongs to the domain of actually instantiated states, which is described by the geometry of general relativity and which exhibits the irreversible ordering that is the subject of the present paper. The failure of quantum mechanics and general relativity to unify—the central problem of modern theoretical physics—is, on this analysis, the consequence of treating the description of the possibility space as if it were the description of the realisation space. The temporal arrow's refusal to fit inside four-dimensional geometry is the same disease presenting in a different organ: in both cases, a description-level property (the group structure of $U(t)$; the two-directional t -axis) is being mistaken for an ontological property of the physical process (the irreversible ordering of realised states).

Third: the semigroup structure of the realisation domain emerges from within standard theory whenever the theory is forced to describe what actually happened rather than what was possible.

The Lindblad equation, which governs the dynamics of open quantum systems and the process of decoherence, has semigroup structure [10,11]: the dynamical map it generates is not invertible in general, because the system's interaction with its environment involves information transfer to degrees of freedom that are not tracked. This is not a failure of the fundamental theory; it is what fundamental theory produces when it must describe an actual physical process of state realisation rather than an abstract evolution in possibility space. The emergence of semigroup structure in these contexts is not evidence that the fundamental theory is secretly irreversible; it is evidence that whenever physics is forced to describe actuality rather than possibility, the semigroup ordering relation—not the group

structure—is what appears. The semigroup is not imported from outside physics. It is what physics itself produces at the realisation boundary.

This point functions as supporting evidence only, not as the primary argument; the primary argument is the distinction between description-level and realisation-level properties developed above. The semigroup structure of the Lindblad equation confirms the conclusion but the conclusion does not depend on it.

The objection that semigroup structure is a feature of non-fundamental physics is therefore answered on all three fronts: the group structure of $U(t)$ belongs to the description level, not the realisation level; the inverse operator $U(-t)$ is a mathematical possibility that is never physically realised; and even within standard theory, the realisation domain consistently produces semigroup structure at every point where it is forced to describe actuality. The semigroup characterisation of temporal structure is not a smuggling of macroscopic irreversibility into fundamental physics. It is the recognition that fundamental physics, at the realisation boundary, has always been producing semigroup ordering and calling it by other names.

Section 3: The Separation of Coordinate Accessibility from Physical Accessibility

3.1 The Map and the One-Way Street

A standard defence of the four-dimensional spacetime ontology proceeds as follows: coordinate axes do not need to be physically traversable in both directions for the coordinate system to be legitimate. The coordinate r in spherical coordinates ranges from 0 to infinity; negative values of r are not physically meaningful, but this does not disqualify r from being a useful and legitimate coordinate. Similarly, the t -axis in Minkowski spacetime has both positive and negative values, and the negative values simply designate coordinates at earlier times—they are reference positions, not inaccessible territories. The analogy to a map is typically invoked: a map can represent territory that is inaccessible (a cliff face, a restricted military zone) without the representation being illegitimate.

This defence is correct as far as it goes. Coordinate systems do not require bidirectional physical traversability. The map analogy is accurate. But the defence proves too much, and in proving too much, it destroys what it was meant to protect.

The analogy to a map with a one-way street is precisely instructive—but not in the way the defender intends. A one-way street restriction on a map does not come from the map's coordinate system. The map's coordinates place the street on a grid; the restriction on direction comes from the traffic law, which is a rule that exists outside and above the coordinate system. The coordinate system is indifferent to directionality. The directionality constraint is imposed by a structure external to the coordinates.

Apply this observation to the four-dimensional case. If the t -axis in Minkowski spacetime is compared to a map coordinate that happens to have a one-way restriction, then the restriction must come from somewhere outside the coordinate system—from some structure that is not itself the t -axis, not itself the Minkowski metric, not itself any element of the four-dimensional geometric framework. That external structure—whatever it is—is the source of time's directionality. That external structure is what makes the negative temporal direction unavailable. And if that external structure is the real source of temporal directionality, then that external structure, not the t -axis, is what time actually is.

The defence of the t -axis's ontological legitimacy by analogy to map coordinates thus becomes a confession: the four-dimensional geometry does not contain the source of temporal directionality within itself. The moment the defender says " t is like a map coordinate with a one-way restriction," they have granted that the directionality constraint comes from outside the geometry. And the structure that comes from outside the geometry to supply what the geometry cannot—that is the temporal ontology. The t -axis is the map. Time is the traffic law. The map is not the territory, and the coordinate is not the time.

3.2 The Minkowski Metric Signature Is Not an Answer

The most technically careful defence available to the four-dimensional ontologist takes a different form: the Minkowski metric signature, in which the temporal direction carries a negative sign, is not merely a notational convention. It encodes the real physical distinction between timelike and spacelike separations. Light cones are causal structures, not mere coordinate artefacts. The metric signature places genuine physical constraints on which events can causally influence which other events. The asymmetry between time and space is therefore not invisible to the geometry; it is encoded in the sign.

The Minkowski line element is [12]:

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2 \quad (1)$$

The negative coefficient of dt^2 is the signature of the temporal direction. It distinguishes timelike intervals ($ds^2 < 0$) from spacelike intervals ($ds^2 > 0$) and lightlike intervals ($ds^2 = 0$). The light cone at any event divides all other events into the causally accessible future, the causally accessible past, the causal present (on the light cone), and the causally inaccessible spacelike region. This is a real structural feature of Minkowski geometry, and it does real physical work.

The response to this defence requires precision. The Minkowski metric signature accomplishes two things: it distinguishes the timelike direction from the three spacelike directions (by assigning it a different sign), and it divides the timelike direction into two causally distinct classes (future-directed and past-directed timelike vectors). Both of these accomplishments are genuine. Neither of them accomplishes what is needed to vindicate the four-dimensional temporal ontology.

The first accomplishment—distinguishing timelike from spacelike—does not establish that the timelike direction is the same kind of entity as a spatial dimension. It establishes the opposite: the temporal direction has different structural properties from the spatial directions, which is exactly what the present argument claims. The metric signature is the geometry's own acknowledgment that t is not like x , y , or z . It is strange to cite this acknowledgment as evidence that t is like x , y , and z .

The second accomplishment—dividing the timelike direction into future-directed and past-directed—is where the decisive point must be made. The Minkowski light cone structure is geometrically symmetric between future and past. The future light cone and the past light cone are geometrically identical structures, related by the transformation $t \rightarrow -t$. Minkowski geometry is perfectly time-reversal symmetric. Nothing in the metric, nothing in the light cone structure, nothing in the signature of the line element breaks the symmetry between future-directed and past-directed timelike vectors. The metric is as silent about the wall as the t -axis itself.

The causal structure of Minkowski spacetime tells us which events can causally influence which other events; it does not tell us which direction of causal influence is "forward." The distinction between "causally before" and "causally after" is encoded in the causal ordering of events, which is an ordering relation—exactly the kind of structure identified in Section 1.2 as Possibility Three. The metric signature identifies the type of separation between events; the ordering relation assigns the direction. These are different structures. The four-dimensional geometry provides the first; it does not provide the second. The wall—the structural asymmetry between past and future—is not inside the geometry. Four-dimensional spacetime is a framework that is, by its own mathematical structure, blind to the directionality that constitutes the most fundamental property of time.

Section 4: The Negative-Time Dilemma

4.1 The Original Argument and Its Repair

The argument from which the dilemma of this section develops originates from a simple observation about coordinate axes. If the time coordinate t is ontologically parallel to the spatial coordinates x , y ,

and z —if it is a genuine fourth dimension of the same fundamental kind—then its negative direction ought to carry the same ontological weight as the negative directions of the spatial axes. Negative x positions exist; negative y positions exist; negative z positions exist. If t is genuinely a spatial-type dimension, negative t positions ought to exist in the same sense.

One version of this argument—the original version—draws the immediate inference: if negative t positions exist, there must be a physical domain corresponding to them, a "negative-time spacetime" that exists alongside our own. If such a domain does not exist or cannot be shown to exist, then the t -axis has a structural deficiency that disqualifies it from the status of a genuine dimension. The absence of a negative-time domain would reveal that the t -axis is not, after all, a genuine fourth dimension but only a parameterisation of a one-sided extent.

The original version of this argument is not fully rigorous. The inference from "the coordinate axis has negative values" to "there must be a physically real negative-time domain" involves a step that requires additional justification. The negative values of a coordinate axis do not automatically commit one to the existence of physical territory corresponding to those values; this is established by the very examples used by defenders of the four-dimensional view (polar coordinate r , spherical coordinate θ , etc.). The argument as originally stated can be deflected by pointing out that the physical geometry may be defined on only a subset of the mathematical coordinate space, so that the negative t -values are mathematical artefacts rather than indicators of a missing physical domain.

The repair of this argument consists in recognising that the deflection itself constitutes a concession—a concession that generates the dilemma. If the defenders of the four-dimensional ontology accept that the negative half of the t -axis has no physical referent, they have accepted Possibility One from Section 1.2: the negative direction is a boundary-condition artefact, a mathematical overhang with nothing physical behind it. And a coordinate axis of which one half is a mathematical overhang is not the description of a physical dimension; it is a notation for a half-line. The dilemma is not between two versions of the original argument; it is between two horns that the defence itself generates.

4.2 The Dilemma in Its Sharpest Form

First Horn.

Suppose the defenders of the four-dimensional temporal ontology maintain that the t -axis is genuinely parallel to the spatial axes—that it is an ontologically full dimension, with both positive and negative directions carrying equal geometric reality. Then the negative direction of the t -axis carries the same ontological weight as the negative directions of x , y , and z . The negative x -direction is physically real; objects exist in negative- x territory. The negative y -direction is physically real. The negative z -direction is physically real. By exact analogy, the negative t -direction is physically real. But if the negative temporal direction is physically real, then there exists a physically real temporal sector corresponding to the negative orientation of the temporal axis—a domain in which the temporal coordinate takes negative values and which possesses the same structural status as any region in which t is positive. This domain is, in our observable universe, physically inaccessible: no observer in the positive- t domain can reach it, observe it, or interact with it.

If the defenders accept this conclusion, they have committed themselves to the existence of a physically real, causally isolated, observer-inaccessible parallel domain—a negative-time universe alongside our own. This commitment is empirically unbounded: it posits a domain that, by construction, generates no empirical consequences for any observer in our domain. It is not a theoretical prediction that extends beyond current observational reach (as, for example, predictions about regions beyond the observable horizon are); it is a domain that is in principle causally disconnected from all observation. A physical theory that commits to in-principle-inaccessible parallel domains as a consequence of its fundamental ontology has abandoned the constraint of empirical accountability. The defenders of the four-dimensional ontology, if they accept the first horn, accept this consequence. It is a consequence

that standard physical practice—which operates under the methodological principle that physical claims must in principle be capable of generating empirical consequences—cannot absorb.

The first horn kills the four-dimensional temporal ontology because it requires the ontology to be empirically unbounded. It commits the defender to a claim that generates no predictions, no constraints, and no observable consequences, in perpetuity, by definition. That is not physics; it is unconstrained metaphysics.

Second Horn.

Suppose the defenders reject the first horn—as they must, given the commitments of empirical physics—and maintain instead that the negative half of the t -axis has no physical referent. The negative t -values are mathematical extensions of the coordinate system, just as $r < 0$ is a mathematical extension of the radial coordinate that has no physical meaning. On this account, the physical time coordinate is not a full axis from $-\infty$ to $+\infty$ but a ray from some initial time to $+\infty$, with the negative half being purely formal.

This concession immediately generates a problem. A coordinate axis of which one half carries no physical referent is not the description of a physical dimension; it is the description of a physical half-line dressed in dimensional notation. If the defenders of the four-dimensional ontology claim that t is a genuine fourth dimension—ontologically parallel to the three spatial dimensions—they cannot simultaneously hold that one half of the t -axis is physically vacuous. The three spatial dimensions do not have physically vacuous halves: negative x -positions are as physically real as positive x -positions. If the negative half of the t -axis is conceded to be formally vacuous, then t is not genuinely parallel to x , y , and z . It is structurally different from them in precisely the way that the present paper claims: the spatial dimensions are group-structured (bidirectional, with both orientations physically real) and the temporal direction is semigroup-structured (unidirectional, with only one orientation physically real). The second horn concedes the argument.

4.3 Against the Defence: "Parameters Need Not Be Complete"

A final line of defence is available: mathematical parameters need not cover all possible directions. Polar coordinate r ranges only over non-negative values; it is a legitimate and useful coordinate without being "complete." Therefore, the t -axis's one-sidedness does not disqualify it from being a legitimate coordinate—and the ontological conclusion does not follow.

This defence is accepted, completely and immediately. Mathematical parameters need not be complete. The conclusion drawn from incompleteness is not that t is an illegitimate coordinate. The conclusion is that t is a coordinate—and not an ontological dimension. The argument is not " t is illegitimate because it is one-sided"; the argument is " t cannot claim ontological status as a dimension precisely because it is one-sided, and the one-sidedness is conceded."

The case of $r \geq 0$ illustrates the point perfectly. Nobody claims that r is an ontological dimension of the universe. Nobody argues that radial distance is a fundamental dimension of reality in the way that the three Cartesian spatial dimensions are. The r coordinate is acknowledged, without controversy, to be a descriptive tool—a convenient parameterisation for problems with spherical symmetry. Its one-sidedness causes no confusion because no ontological claim is attached to it. It is exactly the status this paper proposes for t : a descriptive tool, a bookkeeping parameter, a convenient indexing device for ordering events along the temporal direction. The r analogy, deployed as a defence of the four-dimensional ontology, is a concession to the present argument: it is the implicit acknowledgment that one-sided parameters are tools, not dimensions. To invoke this concession as a defence of t 's dimensional status is to try to have both the tool exemption from completeness and the ontological status of a dimension. The exemption and the status are incompatible. They cannot be held simultaneously.

Section 5: Thought Experiment Two—The Administrator and the Progress Bar

5.1 Full Statement

A being—call them the player—inhabits a simulated world. The simulation has been described to them as four-dimensional: it has three spatial dimensions and one temporal dimension. The player can move freely in the three spatial dimensions. In the temporal dimension, they can only move forward: the simulation advances through a sequence of states and the player cannot return to a prior state, no matter what they do within the simulation. They cannot explain this asymmetry. They have accepted the four-dimensional description of their world as correct, but the temporal dimension behaves unlike the spatial ones, and they cannot account for the difference from within the resources of their world.

An administrator exists outside the simulation. The administrator has a progress bar—a parameter that indexes the simulation's state sequence. The progress bar displays a range from A (the beginning of the simulation) to Z (the end). Currently, the simulation is at state B, somewhere in the middle of this range. The player is at state B, with all their memories, experiences, and physical states corresponding to B.

The administrator uses an external interface to drag the progress bar from B back to A. The entire simulation is reset: every particle, every field, every physical state, every memory trace, every neural configuration—everything that constitutes the simulation's state—is restored to the values it had at A. The player is now at A, with the memories, experiences, and physical state they had at A. They have no memory of ever having been at B. To the player, A is simply their present moment. The entire sequence from A to B has vanished—not because it was erased, but because the very memory system that would have recorded it has been restored to the state it was in before it began recording.

Has the player experienced time reversal? Have they returned to the past?

5.2 Formal Analysis

The simulation's complete state is represented as:

$$S(t) = \{x(t), M(t), I(t)\} \quad (2)$$

where $x(t)$ is the player's physical state (position, configuration, all physical variables), $M(t)$ is the player's memory state (the stored records of their experience sequence), and $I(t)$ is the world's complete information state (all physical states of the simulated environment).

The sequence of states experienced by the player prior to the administrator's intervention is:

$$S(t_0) \rightarrow S(t_1) \rightarrow S(t_2) \rightarrow \dots \rightarrow S(t_B) \quad (3)$$

where t_0 corresponds to state A and t_B corresponds to state B. The player's memory sequence is:

$$M(t_0) \rightarrow M(t_1) \rightarrow M(t_2) \rightarrow \dots \rightarrow M(t_B) \quad (4)$$

The administrator's operation is the external map λ :

$$\lambda: S(t_B) \rightarrow S(t_0) \quad (5)$$

This map restores the complete state of the simulation, including the memory state M , from its value at t_B to its value at t_0 .

After the operation, the player's state is $S(t_0)$, with memory state $M(t_0)$. The player's memory sequence contains only:

$$M(t_0) \rightarrow M(t_1) \quad (6)$$

(as the simulation advances again from A). The player has no record of $M(t_2)$, ..., $M(t_B)$. There is no trace, no residue, no echo of the A-to-B sequence in the player's current state.

Key lemma.

If the external parameter reversal restores the complete state, including the memory state M , then the internal observer's experience sequence contains no reversal event. The player's experienced sequence

is $M(t_0) \rightarrow M(t_1) \rightarrow \dots$, with no branching, no return, and no trace of any prior sequence. The sequence is indistinguishable from the sequence that would have occurred if no administrative intervention had taken place.

Assertion.

Time reversal, properly defined, requires that the same observer retain historical information from the later state (B) while occupying the earlier state (A). An observer who is at A with the memories of A has not reversed time; they are simply at A. The administrator's operation is not time reversal. It is re-instantiation: the prior state is instantiated again, but no observer experiences the transition from B to A, because the very memory system that would have to carry that experience has been restored to the state it was in before the transition occurred. Re-instantiation and time reversal are physically distinguishable: time reversal would leave a memory record of the B-to-A transition; re-instantiation erases that record by definition.

5.3 Two Levels of Time

The thought experiment reveals a decisive structural separation. The administrator's progress bar is a parameter that ranges over the state sequence of the simulation. It can be set to any value in its range. It is, in the terminology introduced in Section 1, a bookkeeping parameter—an indexing device for the simulation's state sequence. The administrator can manipulate this parameter freely, moving it forward or backward, jumping to any position. The parameter is fully bidirectional and fully reversible. It has group structure: every position has an inverse (moving backward as far as one moved forward returns the bar to its starting position).

The player's internal time—the irreversible ordering of their experienced states, encoded in their memory sequence $M(t_0) \rightarrow M(t_1) \rightarrow M(t_2) \rightarrow \dots$ —is categorically different. It is not a parameter that can be manipulated from outside. It is the sequence itself. When the administrator reverses the progress bar, the player's internal sequence does not reverse; it terminates (in the sense that the restored player does not remember the terminated sequence) and begins again. The player's internal time is not affected by the parameter reversal because the player's internal time is not the parameter.

Central assertion.

Even when the coordinate time parameter is completely reversed, the internal time of any observer does not reverse by one instant. This demonstrates that the reversibility of the coordinate time parameter and the reversibility of time have no intersection. The bidirectional freedom available on the t-axis of four-dimensional geometry has never, at any point, made contact with time itself. The progress bar is not time. The administrator moves the progress bar; time—the irreversible ordering of realised states—is untouched by this operation.

The structural separation between coordinate parameter and internal state sequence carries an immediate consequence for two of the most persistent paradoxes in the philosophy of time. The grandfather paradox assumes that a traveller can return to the past carrying intact the intention and memory required to act there—that the same observer occupies state B with the full informational content of having previously been at state B. The administrator's operation demonstrates that this assumption cannot be satisfied: complete reversal of the coordinate parameter restores the observer's memory state to its prior value, which means the intention to act on the past is erased along with every other record of the intervening sequence. The paradox does not require resolution; it requires the recognition that its defining condition—an observer who remembers the future while occupying the past—is structurally unavailable in any framework that correctly separates coordinate time from internal state succession. The same analysis dissolves the paradox of closed timelike curves. A closed timelike curve is a geometric solution in which a worldline forms a closed loop in the spacetime manifold, allowing a trajectory to return to its own causal past. Even if such a geometric structure exists, an observer traversing it cannot carry forward the memory of having completed the loop, because carrying

that memory would require the internal state sequence to contain a record of its own future—which is precisely the reverse-ordering relation that this paper identifies as structurally undefined. The geometry may close; the internal time sequence cannot. The coordinate closes; the observer does not remember closing it. Both paradoxes dissolve not because they are resolved by a new rule, but because the condition that generates them—a single observer with intact memory traversing a temporal loop—is shown to be structurally impossible once coordinate time and internal temporal succession are correctly distinguished.

The implication is direct and unambiguous. All geometric operations performed on t within the four-dimensional framework—coordinate transformations, time reversal (T-symmetry), parameterisation choices, Lorentz boosts—occur at the level of the progress bar. They are operations on the indexing system. They are not operations on the temporal ordering that constitutes the physical reality of time. Four-dimensional spacetime is, in its entirety with respect to the temporal coordinate, a geometry of the indexing system. It has been mistaken for a geometry of time.

5.4 Proper Time and the Self-Testimony of Relativity

General relativity itself recognises a distinction that, followed to its conclusion, supports the argument of this paper [5]. The theory distinguishes between coordinate time—the t -coordinate in a chosen coordinate system, which depends on the coordinate choice and transforms under coordinate changes—and proper time—the time measured along a specific worldline by an ideal clock travelling with the observer. Proper time is not a coordinate label; it is the physical time of a specific observer, encoded in the invariant interval integrated along the observer's worldline.

This distinction is significant. General relativity, within its own framework, acknowledges that t is a label and that something other than t —the proper time—is the physical time of the observer. The theory itself has separated the coordinate parameter from the physical temporal reality. The coordinate t is, by the theory's own account, a label. The physical time is the proper time.

But this acknowledgment, internal to the theory, creates an unresolved tension at the heart of the block universe interpretation. The block universe interprets the four-dimensional spacetime manifold as the ontological whole of physical reality—a timeless structure in which all events exist simultaneously, the apparent flow of time being an artefact of the observer's local trajectory through the block. The block is constituted in part by the t -coordinate. The t -coordinate is acknowledged by the theory to be a label. The block universe interprets a structure constituted partly by labels as the ontological whole of reality.

This is the crack that already runs through the foundation. This paper does not arrive at the foundation from outside to break it. It points to the crack that is already there and notes that it is structural: the distinction between coordinate time (label) and proper time (physical time) is the distinction between the progress bar and the internal time sequence, in the language of Section 5.2. Relativity itself knows the difference. What it has not done is draw the ontological conclusion that this paper draws: that the label is not the time, that the progress bar is not the clock, and that a manifold constituted partly by labels is not, by virtue of that constitution, an ontological container for physical time.

Section 6: Re-Adjudicating Einstein's "Stubborn Illusion"

In a letter written to the family of his recently deceased friend Michele Besso in March 1955, Albert Einstein wrote: "Now he has departed from this strange world a little ahead of me [13]. That signifies nothing. For those of us who believe in physics, the distinction between past, present and future is only a stubbornly persistent illusion." The sentence is among the most cited in the philosophy of physics, and it has become the canonical expression of the block universe view: the four-dimensional spacetime

manifold exists timelessly, all events are equally real regardless of their temporal position, and the experienced flow from past through present to future is a feature of the observer's local perspective rather than of the structure of reality.

The standard reading of this sentence is that Einstein is denying the objective reality of temporal flow—the "passage" of time, the sense that the present moves through a fixed sequence of moments. On this reading, what Einstein calls an illusion is the experience of time as flowing, as progressing, as having a privileged "now" that distinguishes present from past and future. The physics, on this reading, reveals that the t-axis captures the full structure of time, and that the experienced flow is merely a local observer's perspective on a timeless geometric whole.

This paper adjudicates differently. Einstein's intuition—that something in our ordinary experience of time is an illusion—is correct. The identification of what that something is—is inverted.

Consider what the administrator's thought experiment, developed in Section 5, has established. There are two structures in play. The first is the coordinate parameter t , the progress bar that can be manipulated from outside the system, set to any value in its range, reversed, and jumped. This structure is bidirectional, group-like, and fully reversible. It is exactly the kind of structure that is well-suited to constitute an axis in a geometric manifold. The second is the internal ordering of states experienced by the observer—the irreversible sequence $M(t_0) \rightarrow M(t_1) \rightarrow M(t_2) \rightarrow \dots$ that constitutes the observer's actual experience of temporal progression. This structure is unidirectional, semigroup-like, and irreversible by definition: the observer has no access to the reverse direction, not because a barrier prevents traversal, but because the reverse relation is not defined for the experienced state sequence.

Einstein's "stubborn illusion" targets the wrong structure. The experienced flow of time—the observer's irreversible state sequence—is not the illusion. It is the most robustly real temporal structure available to any physical being: it cannot be manipulated, it cannot be reversed, it is written into the very information structure of the observer's existence. What is the illusion—what is the structure that has been mistaken for what it is not—is the t-axis itself.

The t-axis presents itself as the geometry of time. It sits in the four-dimensional manifold as a coordinate dimension, apparently parallel to the three spatial dimensions, apparently extending in both directions, apparently providing a complete geometric account of the temporal structure of reality. But the analysis of the preceding sections has demonstrated that this appearance is precisely an appearance. The t-axis is a progress bar. It is the external parameter that indexes the state sequence without participating in the state sequence. Its bidirectionality is real, but the bidirectionality is a feature of the indexing system, not a feature of time. Its extension in both positive and negative directions is real, but the negative extension is either physically vacuous (second horn of the dilemma) or committed to an empirically inaccessible parallel domain (first horn). Its incorporation into a Minkowski metric with a distinctive signature genuinely distinguishes timelike from spacelike separations—but it does not, and cannot, generate the directional asymmetry that is time's most fundamental property.

Einstein identified the right target—illusion—and fired at the wrong structure. He fired at temporal flow and hit the physical reality of the observer's irreversible experience. The correct target, visible in retrospect from the analysis of the preceding sections, is the geometric t-axis itself: the image of t as a genuine fourth dimension whose negative direction carries physical reality equal to its positive direction, and whose coordinate structure captures the ontological structure of time. That image is the illusion. It is not stubborn because it is difficult to dislodge from common sense—it is stubborn because it is encoded in the formalism of the most successful physical theory of the twentieth century, and dismantling it requires going below the formalism to examine what the formalism is actually describing.

This paper completes Einstein's diagnosis by reversing its identification: the stubbornly persistent illusion is not temporal flow. It is the t-axis—the image of a coordinate parameter as a genuine ontological dimension of physical reality, with a physically real negative direction equal in status to its positive direction. That image is an illusion in the precise sense relevant here: it is a representation that

has been elevated to an ontology. Einstein's intuition—that physics reveals something about time that ordinary experience conceals—is correct. What physics reveals, properly interpreted, is that the t-axis is not time. The t-axis does not capture temporal flow because the t-axis is not time. Temporal flow—the irreversible ordering of realised states—is the reality. The t-coordinate is the representation that was mistaken for the thing it represents.

Section 7: The Summary Argument—Category Leap and the Demotion of Four-Dimensional Spacetime

7.1 Naming the Central Error

The central error that this paper targets has a name: the category leap from representation to ontology. This is the inferential move in which a mathematical structure that successfully describes a physical phenomenon is thereby assigned the ontological status of the phenomenon it describes. The move is not logically licensed by the mathematics. A description of a thing is not the thing. Predictive accuracy is not ontological identity. A framework that correctly predicts the relational geometry of spacetime events is not thereby equivalent to the physical ontology of space and time.

Four-dimensional spacetime commits this error specifically with respect to the temporal coordinate. The Minkowski framework, and its generalisation in general relativity, successfully characterises the geometric relations among events—the invariant intervals, the causal ordering, the curvature effects on trajectories. These are genuine and important achievements. This paper does not contest them.

What this paper contests is the additional philosophical step: the inference from "t is a successful coordinate in this geometric description" to "time is therefore a geometric dimension of the physical world, ontologically parallel to the three spatial dimensions." This inference is not entailed by the mathematics. It is an additional philosophical assumption that has been treated, in much of the subsequent discussion of four-dimensional spacetime, as if it were part of the physical theory. It is not. It is an ontological interpretation of the theory, and it is false.

The two-part structure of the error is worth making explicit. The first part is the category leap itself: assigning ontological status to a mathematical structure on the basis of its descriptive success. The second part is the specific mismatch that results: the mathematical structure assigned ontological status (a bidirectional, group-structured coordinate axis) does not have the properties of the physical phenomenon it purports to ontologically represent (a unidirectional, semigroup-structured ordering relation). The category leap lands on the wrong category, and the mismatch is structural.

7.2 The Complete Chain of Argument

The argument of this paper proceeds through five stages, each of which establishes a result that the next stage builds upon. The chain is presented here in its complete form, with each link stated as an assertion.

One. Time presents a structural one-way wall: traversal in the negative temporal direction is not merely difficult, or unlikely, or informationally obstructed. It is structurally unavailable. The wall is not a feature of any particular physical theory; it is a feature of the phenomenon that any physical theory of time must account for. (Section 1)

Two. The algebraic structure appropriate to the wall is a semigroup ordering relation, not a group-traversable axis. Spatial dimensions have group structure: both orientations are available, and every displacement has an inverse. The temporal direction has semigroup structure: one orientation is defined, the other is not. These are categorically different structures, not merely quantitatively different degrees

of the same kind of structure. The standard physical theories choose group-structured evolution operators for methodological reasons (computational completeness, analytic tractability), but the properties of the methodological choice cannot be read back onto the object being described. (Section 2)

Three. The Minkowski metric signature is not a substitute for what is missing. The metric signature distinguishes timelike from spacelike separations—a genuine and important structural feature—but it does not break the symmetry between future-directed and past-directed timelike vectors. Minkowski geometry is time-reversal symmetric; the directional asymmetry of time is not a consequence of any element of the geometric structure. The geometry is blind to the wall. (Section 3)

Four. The negative-time dilemma is exhaustive. If the t -axis is a genuine ontological dimension, its negative direction carries equal geometric reality to its positive direction, which commits the ontology to a physically inaccessible negative-time domain. If the negative direction has no physical referent, the t -axis is a half-line dressed in dimensional notation—a representation of a one-sided physical structure—and its claim to ontological dimensional status is structurally disqualified. There is no third horn. (Section 4)

Five. The administrator thought experiment establishes a level separation that is decisive. The coordinate parameter t is the progress bar: reversible, bidirectional, fully manipulable from outside the system. The internal time of any observer—the irreversible ordering of their experienced state sequence—is untouched by any manipulation of the parameter. Complete reversal of the coordinate parameter does not produce one instant of internal time reversal. This demonstrates that the reversibility of t and the reversibility of time are not merely different in degree; they are properties of two different and separable structures. The progress bar is not time. (Section 5)

Therefore. Time is not a dimension. Time is an irreversible ordering relation of states. Four-dimensional spacetime is a successful mathematical representation of event relations; it is not the physical ontology of time. The t -coordinate is a bookkeeping parameter—a progress bar—that indexes events without constituting time. Einstein's t is the progress bar. The mistake is taking the progress bar to be time itself—elevating a coordinate representation to the ontological status of the phenomenon it indexes.

7.3 What This Paper Destroys and What It Preserves

The scope of the argument must be stated with precision, because it is easy to overread and to underread it.

This paper preserves the following, without qualification: the empirical predictions of special and general relativity; the Lorentz transformations and their consequences for the relativity of simultaneity, length contraction, and time dilation; the metric structure of Minkowski spacetime as a description of the geometric relations among events; the geodesic equations of general relativity; the causal structure encoded in the light cone formalism. Every prediction, every experimental result, every application of relativistic physics—all of this is untouched by the present argument.

This paper destroys the following: the ontological status of t as a genuine fourth dimension of physical reality, parallel in kind to the three spatial dimensions; the claim that the four-dimensional spacetime manifold is the ontological container of time, within which time's properties are to be read off from the geometry; the claim that the t -coordinate constitutes a genuine fourth ontological dimension of physical reality, parallel in kind to the three spatial dimensions.

The destruction is not mild. Once t is demoted from a dimension to a parameter, the word "spacetime" loses its ontological content with respect to time. What remains is space—genuinely three-dimensional, group-structured, traversable in both directions along all three axes—and an irreversible ordering relation of states that is not spatial, not geometric, not a dimension. Space exists as a geometric object. Time does not. The phrase "four-dimensional spacetime" remains useful as a description of the mathematical framework; it is false as an ontological claim about the structure of physical reality.

This is not a minor reinterpretation. It is the removal of a claim that has been taken as foundational for more than a century. The removal does not require that the mathematics be discarded. It requires only that the mathematics be identified as what it is: an effective and accurate description, not the thing described.

Section 8: Conditions for Falsification

This paper makes ontological claims, and ontological claims must be falsifiable. The conditions under which the argument of this paper fails are stated explicitly and without hedging. Each condition is stated with precision, and the section of the argument it would overturn is identified. Any one of these conditions, if satisfied, constitutes a refutation of the corresponding element of the present argument.

Falsification Condition One.

If experimental evidence is produced demonstrating that macroscopic time reversal is physically achievable, with the reversal observer retaining their complete historical information from the later state while occupying the earlier state—specifically, if an observer at state A retains full memory of having previously been at state B, and this retention is cross-verifiable through independent records that were not reset—then the semigroup characterisation of temporal structure is falsified and the central argument of this paper fails. Note the precision of this condition: the mere restoration of a prior physical state is not sufficient. The condition requires that the same observer's memory sequence include the record of having traversed both the A-to-B and B-to-A transitions. Re-instantiation, as defined in Section 5.2, does not satisfy this condition, because re-instantiation, by definition, produces no such memory record.

Falsification Condition Two.

If it can be demonstrated, from the geometric structure of four-dimensional spacetime alone --- without introducing any structure external to the geometry, where boundary conditions, entropy gradients, and low-entropy initial conditions all count as external structure --- that temporal unidirectionality follows as a derived result, then Section 3 is falsified and the t -axis regains candidate status as a temporal ontology. The falsifying demonstration must proceed from the Minkowski or Lorentzian (pseudo-Riemannian) metric structure alone and produce the directional asymmetry of time as a derived result without appealing to any auxiliary structure.

Falsification Condition Three.

If a physically accessible negative-time domain is identified—not in the sense of a coordinate re-labelling (which is merely a change of parameter assignment) but in the sense of a physical domain with t -values that are negative relative to our present, accessible to observers in our domain, generating empirical consequences that can be measured and confirmed—then the second horn of the dilemma of Section 4 is removed. Note that this condition requires physical accessibility and empirical consequence: the mere mathematical existence of solutions with $t < 0$ does not satisfy it, since such solutions already

exist within the framework of general relativity and are not interpreted as physically accessible negative-time domains by standard practice.

Falsification Condition Four.

If a physical situation can be constructed in which an internal observer can, by means available to them from within their system, experientially distinguish between "the coordinate parameter was reversed and the complete state was restored" and "no operation occurred"—that is, if the internal observer has some physical means of detecting that they have undergone re-instantiation rather than simply continuing their state sequence—then the level-separation argument of Section 5 is falsified. The condition must be met without the observer appealing to any external record that was not also restored in the re-instantiation operation, since the access to such an external record would mean the re-instantiation was incomplete.

Falsification Condition Five.

If a completed theory of quantum gravity demonstrates that the fundamental temporal structure at the Planck scale is group-structured—specifically, that the basic temporal evolution at this scale is fully invertible in a sense that directly corresponds to the observer-experienced temporal ordering, and not merely to the mathematical structure of the evolution operators—then the semigroup ordering relation claim of this paper is overturned. The condition requires the correspondence with observer-experienced time, because the distinction between operator-level invertibility (which already exists in standard quantum mechanics and does not falsify the present argument) and ontological temporal invertibility is the distinction that the entire argument of Sections 2 and 5 is built upon.

Falsification Condition Six.

If it can be demonstrated that a physically real negative-time domain, as described in Possibility Four of Section 1.2, exists without the ontological cost of positing a second spacetime domain inaccessible to all observers in our domain—that is, if the negative-time domain is shown to generate empirical consequences in our domain that constitute evidence of its physical reality—then the exhaustive argument of Section 1.2 is broken, the four-pronged convergence fails, and the negative-time dilemma of Section 4 is simultaneously undermined at both horns.

These six conditions are all the targets this paper presents to its critics. Satisfying any one of them refutes the corresponding element of the argument. This paper identifies its own vulnerable points precisely because it regards the capacity for precise falsification as what distinguishes a genuine ontological claim from an interpretive preference. The argument stands until one of these conditions is met, and it does not ask for agreement that cannot be retracted.

Section 9: Interface with Contemporary Physics

9.1 Block Universe and Eternalism

The block universe interpretation of four-dimensional spacetime—sometimes called eternalism—is the direct target of the argument in Sections 4, 5, and 6 [14,15]. The block universe holds that all events, past, present, and future, exist with equal ontological status in the four-dimensional manifold, and that the distinction between past, present, and future is an observer-relative phenomenon with no fundamental physical significance. Section 6 has adjudicated directly against this view: the block universe is the progress-bar geometry mistaken for the world.

The additional point that must be made explicit here concerns the relationship between the block universe and the empirical predictions of relativity. Defenders of the block universe frequently cite the

relativity of simultaneity [16,7]—the fact that observers in relative motion disagree about which events are simultaneous—as evidence for the block universe ontology. If simultaneity is relative, the argument goes, there is no privileged "now" and therefore no objective distinction between past, present, and future, which supports the block universe picture in which all events co-exist timelessly.

This argument does not survive the analysis of the preceding sections. The relativity of simultaneity is a feature of the coordinate t —specifically, of the fact that different inertial frames assign different t -values to the same pair of events. It is a feature of the parameterisation. The irreversible ordering relation of states is not a simultaneity relation and is not affected by the relativity of simultaneity. The ordering relation is a structural fact about the sequence of realised states; it holds regardless of which coordinate system assigns which t -value to which event. The block universe inference from relativity of simultaneity to the equal ontological status of all temporal positions conflates the parameterisation-relativity of the t -coordinate with the ontological status of the state-ordering relation. The conflation is, once again, a category leap.

9.2 Emergent Time in Quantum Gravity

Several approaches to quantum gravity—most notably the framework built around the Wheeler-DeWitt equation [17,18,19]—are confronted with what is called the problem of time: in the canonical quantisation of general relativity, the Hamiltonian constraint generates a wave equation in which the time variable does not appear as an external parameter but must be identified with some internal degree of freedom of the gravitational system. The result, in its most radical reading, is a "timeless" fundamental description from which the appearance of time must emerge at some level.

This paper shares the direction of the emergent-time programmes—time is not fundamental in the coordinate sense—but differs decisively in the target of what must be recovered. Most emergent-time proposals aim to recover the coordinate t as an emergent effective quantity [20,21,22]: the goal is to show how a t -like parameter arises from the timeless fundamental structure, so that the successful predictive framework of relativistic physics can be reproduced at the appropriate level of description.

This paper proposes that the recovery target is wrong. What must be recovered from the fundamental structure is not a t -coordinate but the irreversible ordering relation of realised states. The t -coordinate is already a derived, bookkeeping parameter at the classical level; recovering it from a more fundamental structure does not constitute progress toward understanding time. What requires explanation—what any adequate fundamental theory of time must generate—is the ordering relation [23,24]: the fact that states are sequenced, the fact that the sequence is irreversible, and the fact that observers are bound to that sequence in a way that they are not bound to any spatial coordinate.

The emergent-time programmes are on the right track in recognising that t is not fundamental. They are on the wrong track in taking the recovery of t -like structure as their goal. The goal should be the derivation of the ordering relation.

9.3 Causal Set Theory

Of all existing programmes in quantum gravity and fundamental physics, causal set theory [25,26] has the structure most compatible with the conclusions of this paper. Causal set theory takes the causal ordering of events—a partial order relation that captures the "before and after" of events in a manifold-independent way—as the fundamental structure of spacetime. The causal set is a discrete partial order [26,27,28]: a set of elements with a relation that is reflexive, antisymmetric, transitive, and locally finite. The continuous spacetime manifold is, in this programme, an approximation to an underlying discrete causal order [29,30,31].

The partial order relation of causal set theory is, in algebraic terms, closer to the semigroup ordering structure that this paper identifies as the correct characterisation of temporal structure than to the group-structured coordinate axis of the standard four-dimensional framework. A partial order is not in general

invertible: if $a \leq b$ (event a causally precedes event b), it is not the case in general that $b \leq a$. The ordering relation has a direction, and the direction is not reversed by any element of the theory's structure.

This paper does not derive its conclusions from causal set theory, and the argument of the preceding sections is logically independent of causal set theory's axioms. The compatibility between the two is structural, not foundational. What causal set theory demonstrates—and what is relevant here—is that it is possible to build a rigorous and formally developed physical framework in which the fundamental temporal structure is an irreversible ordering relation, not a bidirectional coordinate axis. The existence of this programme shows that the position defended in this paper is not physically homeless: it has a natural ally in the most structurally honest of the existing approaches to quantum gravity.

The specific difference between causal set theory and the present paper is in the scope and methodology of the claim. Causal set theory is a programme for quantum gravity: it aims to derive a complete fundamental physical description from discrete causal sets. This paper makes a more limited claim: that the ontological character of time is correctly represented by an irreversible ordering relation and not by a coordinate axis, and that the four-dimensional spacetime ontology fails because it has elevated a coordinate representation to an ontological role for which it is structurally unfit. The claim is ontological-diagnostic rather than theory-constructive.

9.4 Thermodynamic Time Arrow Research

The relationship between this paper and the extensive literature on thermodynamic time arrows requires explicit delimitation, because failure to draw this boundary clearly is the most likely source of confusion about what the present argument is and is not claiming.

The thermodynamic time arrow literature—from Boltzmann's original analysis through the modern discussions of Penrose's conformal cyclic cosmology [32,33,34,35,2], the Past Hypothesis of Albert and Loewer, and the work of Carroll on the cosmological arrow—addresses the following question: given that the microscopic laws of physics are (largely) time-reversal symmetric, why do we observe macroscopic processes that are overwhelmingly asymmetric in time? Why do eggs break but not spontaneously reassemble? Why does heat flow from hot to cold but not spontaneously from cold to hot? The answer in this literature is broadly entropic: the macroscopic asymmetry is a consequence of the universe's extremely low-entropy initial conditions combined with the statistical behaviour of large numbers of degrees of freedom [1,7,16]. The asymmetry is a macroscopic emergent feature, not a fundamental feature of the microscopic laws.

This paper's argument is about a completely different question. The thermodynamic literature asks: why are macroscopic processes difficult or impossible to reverse? The answer is probabilistic and statistical: the reverse processes are not impossible in principle; they are overwhelmingly improbable in practice. This paper asks: why is the negative temporal direction structurally absent—not improbable, but undefined? The answer is that the ordering relation of temporal states is a semigroup relation, not a group relation: the reverse relation simply does not exist at the structural level. These two questions operate at different levels of the physical description and have different answers.

The thermodynamic literature and the present paper do not compete. The thermodynamic arrow explains macroscopic irreversibility at the statistical level; this paper identifies the structural irreversibility at the ontological level. Both explanations are correct at their respective levels. The distinction is the one drawn in Section 1.1: statistical asymmetry means that the reverse process is improbable; structural asymmetry means that the reverse relation is undefined. Both kinds of asymmetry exist; they require different explanations; neither explanation renders the other unnecessary.

9.5 The General Verdict: A Test, Not an Interpretation

This paper does not offer a new interpretation of time [36,3]. An interpretation of time takes an existing physical framework and proposes a reading of its content. This paper does not propose a reading;

it proposes a test. The test is this: any physical theory that claims to describe time must first answer the question of the algebraic type of the temporal structure it posits. Is the temporal structure group-like (bidirectional, with inverses defined for all temporal displacements) or semigroup-like (unidirectional, with the reverse relation undefined)? If the theory posits group-like temporal structure, it must explain the wall—the structural unavailability of the negative temporal direction—from within its own resources, without importing external structure. If it cannot, its claim to have incorporated time into its framework is not a claim about time; it is a claim about a bidirectional parameter that happens to carry the label "time."

Four-dimensional spacetime fails this test. It posits a bidirectional t-axis, it incorporates the t-axis into a Minkowski metric that distinguishes timelike from spacelike separations, and it cannot derive the directional asymmetry of time from within this structure. The wall stands outside the geometry, unexplained. The four-dimensional framework is therefore not a framework for time; it is a framework for a bidirectional parameter that indexes events along the temporal direction. That framework is successful, useful, and empirically accurate. It is not a framework for time.

9.6 Interface for Subsequent Research

This paper establishes that time is an irreversible ordering relation of states and removes the geometric dimension status of the t-coordinate. It deliberately does not answer a deeper question that the argument opens: where does the ordering relation reside?

The administrator thought experiment of Section 5 has already pointed at this question. When the administrator restores the complete state of the system—including the observer's memory—the observer experiences no reversal, because their irreversible state sequence is indistinguishable from the state sequence that would have occurred without any intervention. The observer's internal time is tied to the continuity of their information state. This suggests—but does not prove—that the irreversibility of temporal ordering is not a property of external physical structures but of the information continuity of observers: that the one-way character of time is the one-way character of memory.

If this suggestion is developed into a thesis, the thesis is: temporal irreversibility is not a property of the universe in itself but a property of systems with memory—systems whose current state encodes the history of their prior states in a way that is not erased by the system's ongoing evolution. For such systems, the ordering of states is irreversible because the memory of prior states is carried forward; there is no reverse because memory does not run backward. For systems without memory, the question of temporal direction may not be defined in the experiential sense.

This thesis—that the arrow of time is the arrow of memory, and that temporal ordering is the ordering of information-preserving states rather than a property of external spacetime geometry—is a complete paper in itself. The present paper does not develop it. The present paper establishes its necessary precondition: that time is not a coordinate axis, that the ordering relation is the temporal reality, and that the question of the ordering relation's residence is therefore the correct question to ask [37]. This paper pushes the door open; the subsequent work walks through it.

Section 10: Conclusion—The Physics on the Progress Bar

A century of theoretical physics has been conducted on the progress bar. The coordinates of four-dimensional spacetime have been taken as the geometry of reality—the frame within which the laws of nature operate, the container in which events are located, the structure from which the properties of time are to be read. This paper has argued that this entire framework, in its ontological self-understanding, rests on a category error: the coordinate t, a bookkeeping parameter for event sequences, has been

elevated to the status of a geometric dimension of physical reality, without the evidence required to support that elevation and against the evidence that the preceding analysis has assembled.

The wall was always there. Every being that has ever tried to return to yesterday has encountered it. It is not a wall of thermodynamic improbability, not a wall of computational complexity, not a wall of energy requirements. It is the wall of structural absence: the negative temporal direction is not a difficult direction to traverse, it is not a direction at all. Physical theory, for more than a century, has chosen to interpret this wall as a boundary condition, as an entropic consequence, as an artefact of coarse-graining—anything except what it actually is: evidence that time does not have the structure of a geometric dimension.

The t-axis has two directions. Time has one. This is not because time is a strange or deficient dimension; it is because time is not a dimension. Time is an ordering relation. Ordering relations have a direction without having two directions, because the relation that constitutes the ordering is defined in one orientation and not the other. This is not a limitation; it is the correct description of what ordering relations are. Space consists of three dimensions, each of which has two directions. Time consists of one ordering relation, which has one direction. These are different structures. Four-dimensional spacetime conflates them by assigning to the ordering relation the notation of a dimension. The notation is powerful, convenient, and mathematically fertile. It is not accurate.

The progress bar is not the game. The administrator who moves the progress bar does not move through time; they move through a parameter. The player inside the game, whose memory builds state upon state in an irreversible sequence, lives in time—not in a four-dimensional block, not in a geometric manifold with a t-axis, but in the sequence itself. The sequence is time. The parameter that indexes the sequence is the progress bar. The physics of the last century has been conducted, with extraordinary mathematical sophistication and experimental accuracy, on the progress bar. The game—the actual temporal structure of physical reality—has not yet been the object of an adequate physical theory.

Time is the ordering. The ordering is the world's refusal to provide a return ticket. The geometry of the progress bar described that refusal with great precision but did not explain it, because no description of the progress bar can explain a property that the progress bar does not have. The explanation of the wall requires a theory of the ordering relation—its structure, its scope, its residence—and not a more refined geometry of the coordinate that has never reached it.

The account books are accurate. The world they record is not yet understood.

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